IGCSE 0607 Practice Questions — E1.5

Ordering

Total Questions: 18 — **Total Marks:** 78 Marks

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1 Insert one of the symbols $<$, $>$ or $=$ to make each statement correct.

(i) $0.45 \ldots\ldots \dfrac{2}{5}$

(ii) $-3 \ldots\cdots -7$

[2]

2 Write the following in order of size, starting with the smallest.

$$\frac{5}{8} \qquad 0.62 \qquad 61.5\% \qquad \frac{3}{5}$$

..... **[2]**

3 Write $<, >$ or $=$ to make each statement true.

(a) $0.45 \ldots\ldots\ldots \frac{4}{9}$

[1]

(b) $2^{-3} \ldots\ldots\ldots 0.125$

[1]

(c) $-1.5 \ldots\ldots \cdots -\frac{7}{5}$

[1]

4 **(a)** Insert one of the symbols $<, =$ or $>$ to make each statement true.

(i) $\frac{5}{8} \ldots 0.65$

[1]

(ii) $72\% \ldots \frac{18}{25}$

[1]

(b) Write the following in order of size, starting with the smallest.

$$0.55 \qquad \frac{6}{11} \qquad 54\% \qquad \frac{5}{9}$$

..... **[2]**

5 **(a)** Write the following numbers in order of size, starting with the smallest.

$$\frac{3}{4} \qquad 0.72 \qquad 74\% \qquad \frac{5}{7} \qquad 0.7\dot{4}$$

..... **[2]**

(b) Complete each statement by inserting the correct symbol from $=$, $>$, or $<$.

(i) $-\frac{5}{8}$ ..... -0.65

[1]

(ii) $(0.3)^2$ 9%

[1]

6 **(a)** Insert $<$, $>$ or $=$ to make each statement correct.

(i)

$$0.4 \ldots\ldots \frac{2}{5}$$

[1]

(ii)

$$-7 \ldots\ldots -9$$

[1]

(iii)

$$2^3 \ldots\ldots \sqrt{25}$$

[1]

(b) Write these in order of size, starting with the smallest.

$$33\% \qquad \frac{1}{3} \qquad 0.3 \qquad \frac{3}{8}$$

..... **[2]**

7 **(a)** Given that $0 < p < 1$, write the following expressions in order of size, starting with the smallest.

$$p \qquad p^2 \qquad \sqrt{p} \qquad \frac{1}{p}$$

..... **[2]**

(b) Given that $-1 < q < 0$, write $<, >$ or $=$ to make each statement true.

(i) $q^3 \ldots\ldots\ldots q$

[1]

(ii) $\frac{1}{q} \ldots\ldots\ldots q$

[1]

8 **(a)** Given that $0 < x < 1$, write the following expressions in order of size, starting with the smallest.

$$x \qquad x^2 \qquad \frac{1}{x} \qquad \sqrt{x}$$

..... **[2]**

(b) Given that $x < -1$, write the following expressions in order of size, starting with the largest.

$$x \qquad x^3 \qquad x^{-2} \qquad \frac{1}{x}$$

..... **[2]**

9 **(a)** Given that $x = -0.4$, write the following in order of size, starting with the smallest.

$$x \qquad x^2 \qquad x^3 \qquad \frac{1}{x}$$

..... **[2]**

(b) Insert the correct symbol ($<, =$, or $>$) to make the statement true.
Explain your reasoning.

$5\sqrt{3} \ldots 4\sqrt{5}$

[2]

(c) Find the integer m that satisfies the inequality

$$\frac{m}{7} < \frac{5}{9} < \frac{m+1}{7}$$

$m =$ ..... **[3]**

10 **(a)** It is given that $0 < p < 1$.
Write the following values in order of size, starting with the smallest.

$$p \qquad \sqrt{p} \qquad p^2 \qquad \frac{1}{p}$$

..... **[2]**

(b) Write the correct symbol $<$, $>$ or $=$ to complete each logical statement.

(i) If $x < 0$, then $x^3 \ldots\ldots x^2$.

[1]

(ii) If $x > 1$, then $x^{-1} \ldots\ldots 1$.

[1]

(c) Given $a = 2^{50}$, $b = 3^{30}$ and $c = 5^{20}$, write a, b and c in order of size, starting with the smallest.
You must show all your working.

..... **[3]**

11 Let x be a real number such that $0 < x < 1$.

(a) Write the following four expressions in order of size, starting with the smallest.

$$x^2 \qquad \sqrt{x} \qquad \frac{1}{x} \qquad x$$

..... **[2]**

(b) Now let k be a negative integer such that $k \leq -2$.
Write the following four expressions in order of size, starting with the smallest.

$$k^3 \qquad \frac{1}{k} \qquad k^2 \qquad -\frac{1}{k^2}$$

..... **[3]**

(c) Find the range of values of n for which $n^2 \leq n$ and $n \neq 0$.
Express your answer as an inequality.

..... **[3]**

12 Write the following in order of size, starting with the smallest.

$$\frac{4}{33} \qquad 0.12 \qquad 12.1\% \qquad \sqrt{0.015}$$

..... **[2]**

13 Write the following masses in order of size, starting with the largest.

5.2×10^{-4} kg 0.04 g 4.5×10^{3} mg 6×10^{-7} tonnes

..... **[3]**

14 **(a)** Here are four numerical expressions.

$a = 3.1 \times 10^3$

$b = \sqrt{9.8 \times 10^6}$

$c = 5^5$

$d = 56^2$

Work out the value of each expression and write them in order of size, starting with the largest.

..... **[3]**

(b) State the integer n such that

$$n < \sqrt{250} < n + 1$$

$n =$ ..... **[1]**

15 **(a)** Write these numbers in order of size, starting with the smallest.

$$5^{-3} \qquad 8.5 \times 10^{-3} \qquad 0.00805 \qquad \frac{1}{120}$$

..... **[2]**

(b) The mass, m kilograms, of a bag of rice is 12 kg, correct to the nearest whole kilogram. Write down the inequality representing the possible values of m.

..... **[2]**

16 **(a)** Write the following numbers in order of size, starting with the smallest.

$$\frac{7}{8} \qquad 86\% \qquad 0.865 \qquad \frac{5}{6}$$

..... **[2]**

(b) Write down a fraction, f, such that $\frac{1}{3} < f < \frac{1}{2}$.

..... **[2]**

17 Consider the five numbers p, q, r, s and t defined below:

$p = 4.8 \times 10^{-3}$

$q = \dfrac{1}{205}$

$r = \sqrt{2.3 \times 10^{-5}}$

$s = 0.485\%$

$t = (0.07)^2$

(a) Write the five numbers p, q, r, s and t in order of magnitude, starting with the largest.

..... **[3]**

(b) The number x satisfies the inequality $r < x \leq p$.
Write down a possible value for x in standard form.

$x =$ ..... **[1]**

(c) Given that $y \neq t$ and $y \geq q$, determine whether the statement $y > s$ is always true, sometimes true, or never true.
Explain your answer.

.....

..... **[2]**

18 **(a)** Given that $x > 0$ and $y < 0$, insert the correct symbol $<$, $>$ or $=$ to make the statement correct.

$x^2y \ldots\ldots 0$

[1]

(b) Write these expressions in order of size, starting with the smallest, when $a = 0.5$.

$$a^2 \qquad a \qquad \sqrt{a} \qquad \frac{1}{a}$$

..... **[2]**

(c) State whether the following statement is True or False.
Give a reason for your answer.
If $p > q$, then $\frac{1}{p} < \frac{1}{q}$ for all non-zero real numbers p and q.

..... **[2]**